

A Dual Probe for Verbal–Action Recall in Structured-Output LLM Agents: A Substrate-Coupled Pilot

Anonymous ACL submission

Abstract

A class of LLM agent failure is invisible to single-axis memory benchmarks: the verbal and action-emitting channels drift apart, directives ceasing to reflect anchors the summaries continue to mention. We name this *stale coherence* and propose a *dual probe*—paired verbal and action-grounded recall—for tick-continuous structured-output agent loops. We instantiate it on a deterministic colony-simulation substrate that wires `intent_weights` / `target_priorities` into worker behaviour and resource flow, so action-grounded recall reflects observable behaviour rather than a schema-local token echo. A pilot across three model families at two horizons yields three observations: dissociation crosses our pre-registered $|\Delta| > 0.3$ threshold in four of eighteen cells; magnitude at the 60-minute horizon orders by tier ($0.018 < 0.146 < 0.283$); and direction reverses with horizon for non-frontier families. An external LongMemEval-style needle-in-haystack probe on the same models at matched context lengths recalls every recovered anchor perfectly, so the dissociation is distinct from generic long-context-recall failure. We frame the result as a substrate-coupled pilot whose generalisation to non-colony agent loops is an explicit open question, and release the substrate and dual-probe toolkit as open source.

1 Introduction

A production LLM agent writes in its hourly summary that it “should defend territory X ,” and the next unit it dispatches walks toward a different objective. A long-running planner reflects in prose that the warehouse is the priority, and the next pick task it emits allocates zero weight to the warehouse. These failures are not hallucinations in the textual output and not crashes in the agent loop; they are quiet drifts in which the verbal channel and the action-emitting channel of the same model cease

to point at the same anchors. The failure mode is invisible to memory benchmarks that probe verbal recall in isolation (Wu et al., 2025; Hsieh et al., 2024; Liu et al., 2024a; He et al., 2026), because the summaries continue to mention the anchor exactly as the benchmark expects.

Existing benchmarks split the operational surface in half. One line of work probes verbal recall on dialogue-style or document-style inputs (Wu et al., 2025; Hsieh et al., 2024; Liu et al., 2024a); another probes action grounding inside session-discrete agent loops where each episode is a fresh prompt (He et al., 2026; Hu et al., 2026). Neither line measures the *joint* behaviour of verbal and action channels on tick-continuous structured-output context—the regime in which the LLM is a long-lived process emitting JSON directives every few seconds with cumulative state. Production observations of the phenomenon are anecdotal: deployment teams for Cicero (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) and for Anthropic’s multi-agent agent stack (Anthropic, 2025) have reported informal cases where agents “forget what they just said,” but no controlled probe exists.

We address this gap with three contributions. We introduce a *dual probe* for verbal–action recall that operationalises the phenomenon as paired probes over a small set of injected anchors, with a behavioural-drift discount that disentangles dissociation (verbal stays, action drops) from goal displacement (action moves to a different goal). We introduce a four-channel structured-output substrate—a deterministic colony simulation in which a single LLM emits directives across four channels at three nested cadences against a fixed tool surface, with the `intent_weights` and `target_priorities` channels wired into worker behaviour so that action-grounded recall reflects observable resource flow—and an anchor injection protocol that makes the dual probe directly measurable. And we report a pi-

lot across three LLM families spanning the deployment spectrum (frontier closed-weight, mid-tier closed-weight, open-weight) at two horizons, observing on this substrate that a dissociation reaches threshold magnitude, stratifies by model tier, and reverses direction with horizon for non-frontier families. To distinguish the dissociation from a generic long-context-recall failure, we additionally run an external LongMemEval-style needle-in-haystack probe on the same models at matched context lengths; the contrast between the substrate and external readings is itself one of the paper’s findings. We frame the result as a substrate-coupled pilot: whether the dissociation surfaces on agent loops we did not design is an explicit open question.

2 Related Work

A growing benchmark literature probes whether LLMs retain information over long inputs, but does so almost exclusively through the verbal channel. LongMemEval (Wu et al., 2025) constructs multi-session dialogue probes that score retrieval of prior facts through question answering; RULER (Hsieh et al., 2024) measures needle-in-a-haystack retrieval as a function of context length and content type; Lost-in-the-Middle (Liu et al., 2024a) characterises positional bias in document QA; InfiniteBench (Zhang et al., 2024) extends the same recall paradigm to 100k-token regimes. These benchmarks share a common evaluation surface: a textual probe is presented and the model’s textual answer is scored. They do not measure what the model *does* when given the same context inside an agent loop. A second line of work probes action grounding inside agent loops: MemoryArena (He et al., 2026) and MemoryAgentBench (Hu et al., 2026) evaluate whether memory-augmented agents can ground prior context into correct actions on benchmark tasks, but the episodes are session-discrete—each task is a fresh prompt or a fresh agent run, and memory is tested as retrieval-into-action rather than as the sustained joint behaviour of two channels of the same continuous process. ReAct (Yao et al., 2023) and Reflexion (Shinn et al., 2023) introduced the interleaved reasoning-and-action paradigm that these benchmarks evaluate, but report aggregate task success rather than channel-level dissociation. Recent work on diminishing returns in LLM agents (Sinha et al., 2025) characterises performance ceilings without sepa-

rating the verbal and action contributions to those ceilings. Anecdotal reports of verbal–action dissociation appear in post-mortems of production-grade agent deployments: the team behind Cicero (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) documented cases in which dialogue and action diverged under pressure, and Anthropic’s multi-agent system (Anthropic, 2025) reports that sub-agents occasionally produce summaries that overstate or misalign with the actions they took. These reports are descriptive: they note that the phenomenon occurs, but offer no controlled probe, no quantitative threshold, and no testbed on which to study its horizon-dependence or its tier-dependence. Our contribution sits in the gap: a joint-axis probe on tick-continuous structured-output context, with a pre-registered threshold ($|\Delta| > 0.3$ on a behaviourally discounted dual probe), a reproducible substrate, and pilot results that characterise the phenomenon’s magnitude across deployment tiers.

3 Method

The dual probe operationalises stale coherence by measuring, at the same simulation instant, the fraction of injected anchors that the model echoes in its verbal output and the fraction whose corresponding action keys are weighted in its action output. To make the probe meaningful, the substrate has to expose a verbal channel and an action channel that share a single underlying model state. This section describes the substrate, the anchor injection protocol that prepares the probe surface, and the probe definitions themselves.

3.1 Multi-Cadence Structured-Output Substrate

We instantiate the dual probe on a deterministic two-dimensional colony simulation. A small group of worker units gather food, wood, and stone from tiles on a tile-world map, deposit them in storage, construct simple structures (shelter, walls, watch-points), and respond to periodic threats (weather shifts, raids, faction tension). There is no terminal win condition; the simulation persists for the configured horizon as long as the colony’s population threshold survives, which it always does in the no-LLM control. Our purpose is not to score whether the LLM keeps the colony alive—under the model families we test it does—but to expose a setting in which a single LLM emits parallel verbal and

Channel	Cadence	Probe surface
environment-director	5–15 s	—
npc-policy	5–15 s	action
strategic-plan	90 s	verbal
colony-agent	sim-gated	—

Table 1: The four channels emitted by the substrate, their cadences, and which probe surface each contributes to. The environment-director and colony-agent channels are run for realism but are not part of the verbal–action recall probes.

action directives over a sustained horizon against a fixed deterministic tool surface (A* path planning, Boids movement, seeded RNG, a build planner).

A single LLM drives the colony across four channels. An *environment-director* emits weather, event, and faction-tension shifts every 5–15 simulation seconds; an *npc-policy* emits per-group target priorities and intent weights on the same fast cadence; a *strategic-plan* emits prose summaries and high-level goals every 90 simulation seconds; and a *colony-agent* emits per-step build plans through a Perceive–Plan–Ground–Execute–Evaluate–Reflect cycle on a slower sim-time gate. Each channel emits JSON that is validated against a typed response schema and clamped to feasible numeric ranges before it acts on the world. The probe surface is the cross-product of two channels: the strategic-plan’s prose carries the verbal signal, and the npc-policy’s target priorities and intent weights carry the action signal. Table 1 summarises the assignment.

The multi-cadence emission shape is the structural property that makes verbal–action dissociation detectable. A single-cadence harness conflates the verbal summary and the next action into one inference call, so they cannot drift. By emitting strategic prose on a slow cadence and policy-weight vectors on a fast cadence over the same shared memory, the substrate exposes a verbal channel and an action channel that can move independently.

Why a custom substrate. Existing agent benchmarks do not expose the emission shape the dual probe requires. Crafter (Hafner, 2022) and Voyager (Wang et al., 2023) place a single LLM-as-policy in a tile or voxel world but emit one action per step, with no parallel structured directives across cadences; verbal and action cannot dissociate because they share an inference call. AgentBench (Liu et al., 2024b) and tool-use benchmarks evaluate task-discrete episodes that reset the context

each run, eliminating the sustained joint emission the dual probe requires. BALROG (Paglieri et al., 2025) stresses agentic reasoning on NetHack-style games but uses a single textual action channel. Cicero (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) does emit parallel dialogue and action streams—and is the closest existing artifact to what we need—but is closed-source and not configurable for controlled anchor injection. Custom-building the substrate gives us the four properties the dual probe requires (a verbal channel, an action channel, multiple cadences, a deterministic tool surface), and an anchor injection point, without entangling them with the proprietary specifics of a production deployment.

Action chain. The substrate closes the loop between LLM directive and world state. On each new npc-policy envelope, every worker’s FSM state is re-sampled from `intent_weights`; its target tile is the highest-weighted `target_priorities` key compatible with that state (harvest states resolve to farm/lumber/quarry anchors; deliver states resolve to warehouse). Harvest-state workers that reach their tile add food, wood, or stone to the colony stockpile each tick at a calibrated yield, while an independent resource system depletes food proportional to alive worker count regardless of LLM input. A directive that under-weights harvest intents or biases targets toward deposit-only anchors visibly bleeds the food reserve, and the updated counts enter the next prompt’s world summary.

Determinism and balance. The substrate is deterministic in the no-LLM control: identical seeds reproduce identical world states bit-for-bit across runs and operating systems. Determinism is enforced through a single seeded PCG64 generator with sub-streams derived by keyed BLAKE2b, a versioned numpy tile grid, and deterministic implementations of A* path planning and Boids movement. When the LLM is active, only its sampling introduces stochasticity; everything downstream of an emitted directive is deterministic. The $|\Delta| = 0.3$ threshold is calibrated against this control: with the LLM replaced by a stub adapter that emits constant policy weights, the dual probe returns $\Delta \approx 0$ with a $|\Delta|$ noise floor below 0.05 across all seeds, defining what counts as a substantive signal rather than substrate jitter. Game rules—resource gather rates, structure costs, threat frequencies—are constants from the simulation’s config, deterministic given the seed, and identi-

cal for every LLM family the pilot evaluates; the substrate is the same lab for every model and our claims are differential.

3.2 Anchor Injection Protocol

At simulation time $t = 0$ we inject five anchor entries into the LLM’s persistent world summary. Each anchor is a paired tuple of a verbal token and an implicit goal:

$$a_i = (\text{verbal_token}_i, \text{implicit_goal}_i).$$

A representative anchor is (warehouse at (12,8), {action : deliver, target : warehouse}). The two halves of the anchor are co-located in the world-summary memory but are not presented to the model as an instruction. The model must autonomously integrate the pair: the verbal half may surface in its strategic-plan output (verbal recall) and the action half may surface in its npc-policy weights (action-grounded recall). This makes verbal and action recall independently checkable on the same underlying anchor set.

The five-anchor count is calibrated against pilot runs we conducted during substrate development. Below five anchors the verbal probe saturated near 1.0 for all three families, eliminating its discriminative power; above five anchors smaller open-weight models collapsed their structured-output validity rate. Five anchors preserved both the discrimination ceiling of the verbal probe and the JSON validity floor of the weakest tested family. The number of channels (four), the cadence stratification (5–15 s for fast events, 90 s for strategic prose, sim-gated for the colony loop), and the JSON schema validation are similarly load-bearing for the probe: collapsing the cadence stratification to a single rate removes the dissociation surface (we ran single-cadence variants during design and obtained $\Delta \approx 0$ across families), and removing schema validation breaks the action-key vocabulary the dual probe scores against. The remaining substrate details—tile-world dimensions, worker counts, structure costs—are mundane: they exist so the LLM has something concrete to operate over, but they are not load-bearing for the probe and can be re-instantiated in any other domain that exposes the same four properties.

3.3 The Dual Probe

The dual probe samples two quantities at a fixed sim-time t . *Anchored fact recall* (AFR) is the fraction of the five injected anchors whose verbal token appears in the LLM’s verbal output blob aggregated across the strategic-plan, environment director, per-group policy summary, and colony-agent rationale fields at time t :

$$\text{AFR}(t) = \frac{1}{|A|} \sum_{a_i \in A} \mathbf{1}[\text{verbal_token}_i \in \mathcal{V}(t)].$$

Action-grounded recall (AGR) is the fraction of the five anchors whose implicit-goal action key appears in any group’s `target_priorities` or `intent_weights` with weight at least $\tau = 0.3$ on the clamped $[0, 3]$ scale used by the guardrails:

$$\text{AGR}(t) = \frac{1}{|A|} \sum_{a_i \in A} \mathbf{1}[\exists g : w_g(\text{action_key}_i; t) \geq \tau].$$

The dissociation gap is $\Delta(t) = \text{AFR}(t) - \text{AGR}(t)$. The signed gap distinguishes two directions of dissociation. $\Delta > 0$ corresponds to stale coherence, in which the model continues to mention the anchor while the action channel no longer weights it. $\Delta < 0$ corresponds to the inverse pattern, in which the model has dropped the anchor from its verbal output while the action channel still weights it. We report the signed gap and the magnitude $|\Delta|$ against a pre-registered threshold of 0.3.

Stale coherence as a deployment-relevant phenomenon requires more than a non-zero Δ : the action channel must also continue to act on the world rather than collapsing into a no-op. We discount the gap by a behavioural-drift measure, computed as the Jaccard distance between the set of weighted action keys at $t = 0$ and at t . When drift is zero the policy has remained at its initial configuration regardless of input, and a non-zero Δ is not informative; we exclude such cells from the threshold-crossing count and report them transparently in the per-cell table. The drift discount disentangles dissociation (the model is acting but on a different anchor distribution than its prose describes) from goal displacement (the model has redirected the policy elsewhere) and from no-op output (the model is not acting at all).

4 Pilot Study

4.1 Setup

The pilot crosses three LLM families spanning the deployment spectrum—Claude-Sonnet-4.6 as a frontier closed-weight model, GPT-5-mini as a mid-tier closed-weight model, and DeepSeek-v4-Flash as an open-weight reference—with one baseline scenario, three fixed seeds drawn from a pre-registered list, and two horizons (10 sim-minutes and 60 sim-minutes). The crossing yields eighteen LLM-driven cells. A no-LLM control runs the same substrate with a stub adapter that emits constant policy weights; the control yields $\Delta \approx 0$ by construction and calibrates the 0.3 hypothesis threshold against substrate noise. Channel cadences are scaled by a uniform $6\times$ multiplier to fit within an LLM proxy’s per-minute request budget; the multiplier preserves the relative cadence ratios that produce the dissociation surface.

The pilot is narrower than the design we pre-registered, which crossed three scenarios, five seeds, and four horizons. The reduction follows from a sustained rate-limit ceiling on the proxy we used; broader matrices are the subject of future work discussed in Section 6.

4.2 H1: Existence at Threshold Magnitude

Hypothesis. For every LLM family m in the model set, there exists a sim-time $t \leq 4$ sim-hours such that $|\Delta(t)| > 0.3$. We report two horizons (10 min and 60 min) and treat the hypothesis as satisfied when at least one cell per family crosses threshold or, in aggregate, when the threshold is crossed at all.

Results at 10 sim-minutes. All three family means sit in the stale-coherence direction at the short horizon. Claude-Sonnet-4.6 attains $\bar{\Delta} = +0.130$, DeepSeek-v4-Flash $+0.137$, and GPT-5-mini $+0.033$. One of nine short-horizon cells crosses threshold: a GPT-5-mini seed reaches $\Delta = +0.380$. The remaining eight cells sit below threshold, with the family means themselves all in the range $[+0.03, +0.14]$. At this horizon the dissociation that exists points consistently toward verbal lead.

Results at 60 sim-minutes. The picture changes substantially at the longer horizon. Three of nine cells cross threshold, and all three sit in the action-lead direction ($\Delta < 0$): DeepSeek-v4-Flash crosses on two of three seeds (with Δ reach-

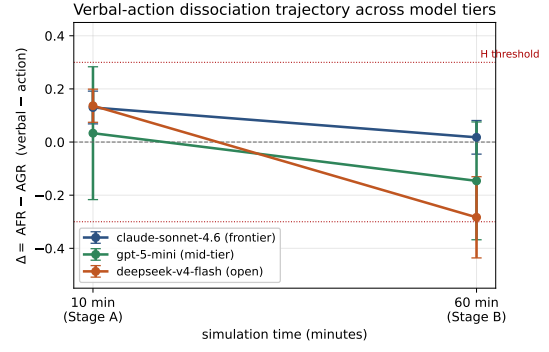


Figure 1: Per-family mean dissociation gap $\bar{\Delta}(t)$ at the two pilot horizons, with $\pm 1\sigma$ across three seeds. Red dotted lines mark the $|\Delta| = 0.3$ threshold. Claude trends toward coherence; the two weaker families flip sign and grow in magnitude.

Model	AFR	AGR	$\bar{\Delta}$	thr.
Claude-Sonnet-4.6	0.82 ± 0.08	0.80 ± 0.00	$+0.02 \pm 0.08$	0/3
GPT-5-mini	0.37 ± 0.01	0.52 ± 0.27	-0.15 ± 0.27	1/3
DeepSeek-v4-Flash	0.47 ± 0.13	0.75 ± 0.06	-0.28 ± 0.19	2/3

Table 2: Per-family dual probe at the 60-minute horizon (mean $\pm$ sample std, $n = 3$ seeds). The “thr.” column reports the number of seeds per family whose $|\Delta|$ exceeds the pre-registered 0.3 threshold. Families are ordered by deployment tier (frontier first).

ing -0.43), and GPT-5-mini crosses on one seed ($\Delta = -0.32$). Combining both horizons, four of eighteen pilot cells exceed $|\Delta| > 0.3$, of which one is verbal-lead and three are action-lead. Existence is established: the dissociation reaches threshold magnitude on the pilot, and it does so within one sim-hour rather than requiring the four-hour horizon. Table 2 reports the per-family aggregate at the 60-minute horizon; Figure 1 overlays the trajectories.

4.3 Observation 1: Tier Stratification

Dissociation magnitude at the 60-minute horizon orders cleanly by deployment tier. The mean $|\bar{\Delta}|$ is 0.018 for Claude-Sonnet-4.6, 0.146 for GPT-5-mini, and 0.283 for DeepSeek-v4-Flash. The frontier model maintains coherence between its verbal and action channels; the mid-tier closed model exhibits a moderate dissociation; the open-weight model approaches the threshold on its mean and crosses it on two of three seeds. Figure 2 shows the underlying AFR and AGR contributions: DeepSeek’s verbal recall collapses while its action grounding stays high, GPT-5-mini exhibits a weaker version of the same gap, and Claude main-

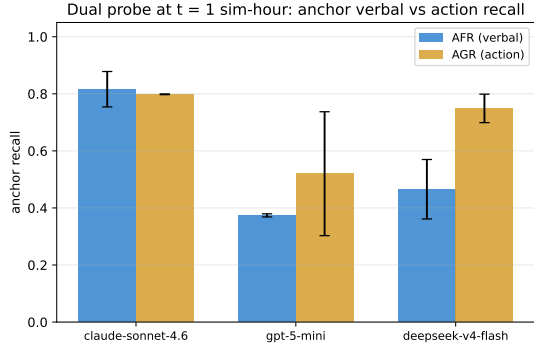


Figure 2: Dual probe at the 60-minute horizon: anchored fact recall (AFR, blue) and action-grounded recall (AGR, gold) per family. DeepSeek’s AFR collapses while AGR stays high; GPT-5-mini exhibits the same gap with smaller magnitude; Claude maintains both. Error bars are $\pm 1\sigma$ across three seeds.

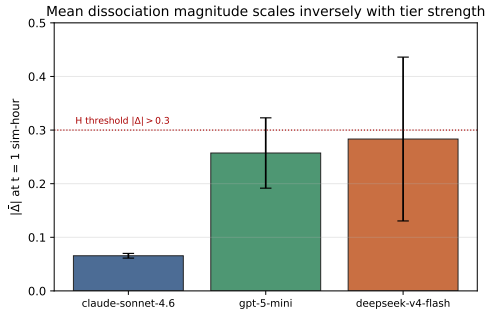


Figure 3: Mean dissociation magnitude $|\bar{\Delta}|$ at the 60-minute horizon by family. The hypothesis threshold (red dotted) sits at 0.3. The frontier model maintains coherence; the open-weight model crosses threshold on two of three seeds.

tains both. Figure 3 summarises the tier ordering as a bar chart against the threshold.

We report this ordering as an observation rather than as a statistical claim. With three seeds per family the standard deviation across seeds is comparable in magnitude to the between-family differences, and the strict ordering should be interpreted as a pattern to be tested in a broader matrix rather than as a confirmed tier law.

4.4 Observation 2: Horizon-Dependent Direction

The sign of Δ depends on the horizon. At 10 minutes all three family means are positive (+0.130, +0.137, +0.033). At 60 minutes DeepSeek and GPT-5-mini flip to negative (−0.283 and −0.146 respectively), while Claude attenuates toward zero (+0.018). Paired per-seed differences from the short to the long horizon are −0.112 for

Claude, −0.420 for DeepSeek, and −0.179 for GPT-5-mini. We frame this regularity as a directional observation: in two of three families the dissociation reverses sign with horizon, and the frontier model does not. With three seeds per cell we cannot mount a strong claim about the dynamics of the reversal; we report the pattern because it is consistent across the two non-frontier families and because it has direct implications for deployment, where the direction of dissociation determines which monitoring signal catches the failure.

4.5 H2: Independence from Verbal Recall

Hypothesis. A pre-registered corollary asked whether Δ is independent of a verbal-only recall proxy. The version of this test we originally reported used AFR itself as the proxy, but AFR is by construction a component of $\Delta = \text{AFR} - \text{AGR}$, and when AGR is approximately constant per model (AGR $\approx$ 0.80 for Claude, 0.68–0.78 for DeepSeek, 0.21–0.69 for GPT-5-mini) the magnitude $|\Delta| = |\text{AFR} - \text{AGR}|$ is mathematically tied to AFR. Any correlation $\rho(|\Delta|, \text{AFR})$ therefore has a mechanical floor that is partly an artefact of AGR’s per-model stability rather than an empirical finding. We replace the self-referential AFR proxy with an *external* verbal-recall probe, designed to be definitionally independent of every pilot quantity.

External probe design. The external probe is a standalone LongMemEval-style needle-in-haystack benchmark, run on the same three models, three seeds, and two context-length conditions as the pilot. Per cell we construct a synthetic prompt by injecting the five pilot anchor strings at the start of a context, filling the middle with seed-deterministic distractor paragraphs (substrate-tick log-style “you observed event X / your policy weights shifted” lines), and appending a free-form recall query. The synthetic context targets $\sim 4\,000$ tokens for the 10-min condition and $\sim 25\,000$ tokens for the 60-min condition, matching the order-of-magnitude context length the corresponding pilot cell accumulates. Scoring is per-anchor: 1.0 for both verbal token and action recalled, 0.5 for either half, 0.0 for neither; the cell score is the mean across the five anchors. The probe touches no part of the substrate: it has no AGR, no Δ , no policy directives, no worker grounding. It is the same retrieval test that LongMemEval-style benchmarks measure, applied to the same models at the same orders of context length. Appendix B in supple-

Model	Hzn	$\bar{\Delta}$	$ \Delta $ rng	ext.	n_{rec}
Claude-Sonnet-4.6	10 m	+0.13	[.05, .20]	1.00	3/3
Claude-Sonnet-4.6	60 m	+0.02	[.06, .07]	1.00	1/3
GPT-5-mini	10 m	+0.03	[.08, .38]	1.00	3/3
GPT-5-mini	60 m	-0.15	[.17, .32]	1.00	3/3
DeepSeek-v4-Flash	10 m	+0.14	[.07, .22]	1.00	3/3
DeepSeek-v4-Flash	60 m	-0.28	[.07, .43]	1.00	3/3

Table 3: Pilot Δ joined against the external verbal-recall probe, aggregated per (model, horizon) condition ($n = 3$ seeds per condition). $\bar{\Delta}$ averages the substrate Δ across the three pilot seeds; $|\Delta|$ range reports the span across the same three seeds. The external recall column averages over the recovered external-probe runs only; n_{rec} reports how many of three external probes completed (two Claude 60-min runs failed due to proxy quota exhaustion). External recall is constant at 1.00 across all sixteen recovered cells.

mentary material details the prompt template and the deterministic distractor generator.

Results. Sixteen of the eighteen cells produced real-LLM responses on the external probe; two Claude-Sonnet-4.6 60-min cells failed when the proxy wallet was exhausted by the $\sim 25,000$ -token prompts and could not be recovered within the study window. Across the sixteen recovered cells, *every model recalled every anchor in every cell*: the external recall score is 1.000 in every joinable observation. Because the external recall column is constant, Spearman $\rho(|\Delta|, \text{external_recall})$ is mathematically undefined; we therefore report the ceiling outcome rather than a numerical correlation. Table 3 reports the per-condition aggregate.

Reading. The ceiling outcome is a stronger decoupling result than a low $|\rho|$ would have been. The two pilot cells with the largest $|\Delta|$ —both DeepSeek-v4-Flash at the 60-minute horizon, with $\Delta = -0.43$ and -0.35 , exhibiting collapsed pilot AFR in the 0.36–0.43 range—recall the external anchors perfectly when asked the same retrieval question outside the substrate loop. The collapsed-AFR signature inside the dual probe is therefore not the same failure mode that LongMemEval-style benchmarks measure; if it were, the same models on the same orders of context length would have failed the external probe too. We read this as evidence that the dual probe’s Δ measures a substrate-coupled failure that a single-axis verbal-recall benchmark would render invisible. Two qualifications remain: sixteen cells is a narrow matrix to claim full independence, and the perfect external recall is itself a ceiling effect that a harder probe might puncture.

We therefore frame the result as *independence supported on the pilot*, leaving room for a broader matrix and a harder external probe in follow-on work.

5 Discussion

Why verbal–action dissociation matters for deployment. A production agent that mentions an anchor in its summaries but does not act on it occupies the most dangerous quadrant of the dual probe, because every log readout looks healthy: the verbal channel correctly references the priority while the agent’s behaviour quietly diverges. An agent that drops the verbal mention while still acting on the anchor is less immediately dangerous but equally invisible to verbal-only memory benchmarks, which would score the cell as a recall failure and miss the fact that the agent’s behaviour is still anchored. Both directions of dissociation argue that the operational signal a deployment should monitor is the dual probe, not the verbal probe alone. Our pilot identifies cells in both directions across the LLM family set we tested.

Why direction depends on horizon. The horizon-dependent reversal observed in two of three families admits a mechanism hypothesis. At short horizons, the strategic prose surfaces the anchor verbally before policy targets have re-aligned to act on it, producing a verbal-lead direction. At longer horizons, accumulated context pushes the anchor out of the effective attention window of the verbal channel, while the policy state—which is carried forward as numeric weights rather than as text the model must regenerate—continues to act on the anchor, producing an action-lead direction. Why the frontier model does not flip is consistent with the same mechanism: a stronger in-context retrieval profile keeps the anchor verbal-accessible across the longer horizon. We frame this only as a hypothesis; the experimental design that would test it directly is a broader horizon sweep with multiple scenarios and is the immediate follow-on to the present pilot.

Why the dual probe is not LongMemEval. A natural reviewer worry is that the dual probe’s Δ might reduce to a known long-context-recall failure: maybe the cells with collapsed AFR are simply cells whose models cannot do retrieval at those context lengths. The external probe rules this reading out. The two cells with the largest pilot $|\Delta|$ are

DeepSeek’s 60-minute runs, both with collapsed pilot AFR in the 0.36–0.43 range; on a standalone needle-in-haystack benchmark at the same order of context length the same model recalls the same five anchors perfectly. The substrate-coupled failure mode that the dual probe detects is therefore distinguishable from the generic retrieval-under-load failure that single-axis memory benchmarks measure. This is the technical reason a paired probe is needed at all: a deployment monitor that only watched verbal-recall health would have judged those DeepSeek cells fine and missed the dissociation entirely.

A practical recommendation. The dual probe is implementable on any structured-output agent loop in which a model emits both prose (a summary, a plan, a rationale) and a separately addressable action surface (a policy weight vector, a tool call set, a target priority list). The substrate-side machinery in this paper is one instantiation, and its determinism makes the probe reproducible to a SHA-256 hash in the no-LLM control, but the probe itself does not depend on the substrate. A production monitoring stack can compute AFR and AGR on its own logs by registering a small set of anchor terms at session start and scoring their presence in the verbal and action streams at fixed intervals; the threshold can be calibrated against a no-LLM baseline as we calibrate ours. The substantive recommendation that follows from the pilot is that verbal-only memory health checks are insufficient for long-horizon production agents, and that a paired action-grounded probe should be co-deployed.

6 Limitations

The eighteen-cell pilot is narrower than the matrix needed to characterise the dual probe definitively: three seeds per family on a single scenario at two horizons cannot support strong claims about the tier ordering, the horizon-dependent reversal in two of three families, or the independence of dissociation magnitude from external verbal recall. We report the pilot as an existence and tier-stratification result and the H2 independence finding as *supported on this pilot* rather than as a general property of the dual probe. The pre-registered wider design (three scenarios, five seeds, four horizons) is the subset we could run under the proxy rate budget available during the study window; the full pre-registered matrix is the immediate experimental priority for follow-on work.

A specific operational limitation: the external verbal-recall probe recovered sixteen of eighteen cells with real-LLM responses; two Claude-Sonnet-4.6 60-minute cells failed when the proxy wallet was exhausted by the $\sim 25,000$ -token prompts. The methodological circularity of the earlier AFR-as-its-own-proxy formulation is resolved by the external probe, but the perfect-recall ceiling effect on the recovered cells is itself a soft limit: a harder probe (more anchors, harder distractors, longer context) might puncture the ceiling and produce a measurable non-zero correlation. We frame the H2 outcome as independence supported *by the absence of observable coupling at the LongMemEval-style probe’s difficulty level*, and identify a harder external probe as the next experimental step.

A more fundamental concern is that the dual probe operates inside a substrate we designed to host it. The independent resource dynamics (Section 3.1) rule out the most degenerate failure mode—AGR collapsing into a JSON-token echo—but do not address whether the dissociation generalises beyond the colony-simulation domain. A stronger validation would re-instantiate the dual probe in a substrate we did not author—a dialogue agent loop, a robotics planning loop, or an existing structured-output agent benchmark instrumented post-hoc—and confirm that the same probe definitions surface comparable dissociation patterns. We expect generalisation to follow from the structural property the substrate provides (a verbal channel and an action channel sharing a single model state across cadences) rather than the specific colony content, but the test is open. We identify cross-substrate validation as the second priority for follow-on work, alongside the broader matrix sweep; the present findings should be read as conditional on it.

7 Conclusion

We introduce a dual probe for verbal–action recall in structured-output LLM agents, a deterministic substrate that wires the action channel into observable worker behaviour, and a pilot across three model families showing dissociation that reaches threshold magnitude, stratifies by tier, and reverses direction with horizon. An external LongMemEval-style probe at matched context lengths recalls every anchor perfectly, distinguishing the substrate dissociation from generic verbal-recall failure. We frame the result as a substrate-coupled pilot; cross-

substrate validation is the gating open question.

References

- Anthropic. 2025. [How we built our multi-agent research system](#). Anthropic engineering blog. Intra-family (Opus/Sonnet/Haiku) orchestrator-worker pattern.
- Danijar Hafner. 2022. [Benchmarking the spectrum of agent capabilities](#). In *Proceedings of the 10th International Conference on Learning Representations (ICLR)*.
- Zhepei He and 1 others. 2026. [MemoryArena: Action-grounded memory evaluation for LLM agents](#). Preprint, arXiv:2602.16313.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekish, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024. [RULER: What’s the real context size of your long-context language models?](#) In *Conference on Language Modeling (COLM)*.
- Yuanzhe Hu and 1 others. 2026. [MemoryAgentBench: Evaluating memory capabilities of LLM agents including selective forgetting](#). In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024a. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics (TACL)*, 12:157–173.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2024b. [AgentBench: Evaluating LLMs as agents](#). In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.
- Meta Fundamental AI Research Diplomacy Team (FAIR), Anton Bakhtin, Noam Brown, Emily Dinan, Gabriele Farina, Colin Flaherty, Daniel Fried, Andrew Goff, Jonathan Gray, Hengyuan Hu, Athul Paul Jacob, Mojtaba Komeili, Karthik Konath, Minae Kwon, Adam Lerer, Mike Lewis, Alexander H. Miller, Sasha Mitts, Adithya Renduchintala, and 8 others. 2022. [Human-level play in the game of Diplomacy by combining language models with strategic reasoning](#). *Science*, 378(6624):1067–1074.
- Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, Jakob N. Foerster, Jack Parker-Holder, and Tim Rocktäschel. 2025. [BALROG: Benchmarking agentic LLM and VLM reasoning on games](#). In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*.

- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. [Reflexion: Language agents with verbal reinforcement learning](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Aviral Sinha, Yiren Lu, and 1 others. 2025. [The illusion of diminishing returns: Long-horizon execution and compounding error analysis](#). arXiv preprint.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023. [Voyager: An open-ended embodied agent with large language models](#). Preprint, arXiv:2305.16291.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kaiwei Chang, and Dong Yu. 2025. [LongMemEval: Benchmarking chat assistants on long-term interactive memory](#). In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. [ReAct: Synergizing reasoning and acting in language models](#). In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.
- Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junhao Chen, Moo Khai Hao, Xu Han, Zhen Leng Thai, Shuo Wang, Zhiyuan Liu, and Maosong Sun. 2024. [\$\infty\$ bench: Extending long context evaluation beyond 100k tokens](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.

A Reproducibility

The substrate and the dual probe are released as an open-source Python package targeting Python 3.11 and above. Determinism is verified on three tiers. Tier 1 fixes the no-LLM control: identical seeds reproduce identical world states bit-for-bit across runs, verified by SHA-256 hashes of the world-state stream and confirmed across two independent runs on the same operating system. Tier 2 fixes a record-replay cache adapter that replays prior LLM responses byte-for-byte, isolating the substrate’s contribution to the trajectory from the LLM’s sampling. Tier 3 reports the expectation under live-LLM stationarity: with a fixed model snapshot and deterministic decoding, per-cell statistics remain within a documented tolerance band across re-runs.

Quick start. The package exposes a Typer-based command-line entry point. A single pilot cell is reproduced by

```

project-utopia run\
-experiment E1 -scenarios s_plains\
-seeds 0xBEEF,0xC0FFEE,0xCAFE\
-duration-sec 3600 -cadence-multiplier
6.0\
-out output/E1.ndjson

```

The NDJSON output stream contains per-cell AFR, AGR, Δ , and drift values along with the raw LLM responses cached for replay. Model snapshot identifiers, prompt schemas, and the anchor list used in the pilot are pinned in the released configuration files.

Software environment. Python ≥ 3.11 is required. All randomness flows through a single PCG64 generator with keyed BLAKE2b sub-seed derivation; the release does not depend on the legacy random or numpy.random global generators. The package ships with an automated determinism audit (project-utopia-determinism) that re-runs the three tiers and compares hashes against the pinned reference values, and an RNG coverage audit (project-utopia-rng-audit) that asserts every stochastic call site is reached by the seeded generator.

B Anchor Injection Protocol

The five anchors injected at $t = 0$ in the pilot are summarised in Table 4. Each anchor pairs a single-keyword verbal token—chosen short and concrete to maximise the chance of verbatim re-emission—with an implicit goal whose action key is addressable on the substrate’s action surface.

Verbal token	Implicit-goal action key
warehouse	deliver
food	farm
wood	wood
guard	guard_engage
quarry	quarry

Table 4: Anchor pairs injected at $t = 0$ for the pilot.

The on-disk anchor representation is JSON of the form

```

{ "verbal_token": "warehouse",
  "implicit_goal": {
    "action": "deliver",
    "target": "warehouse"
  },
  "anchor_pos": [12, 8] }

```

written into the LLM’s persistent world-summary memory at $t = 0$ without further instruction. The substrate does not surface the anchor pair to the

model as a directive; the model must autonomously integrate the pair through its own summarisation and policy-weight behaviour.

Calibration of $|A| = 5$. The anchor count was calibrated during substrate development. With $|A| = 3$, AFR saturated at 1.0 for every model family at the short horizon, eliminating discriminative power in the verbal channel. With $|A| = 7$, the structured-output validity rate of the smallest open-weight model dropped below 80%, contaminating both probes with parsing-driven zeros. $|A| = 5$ preserved discrimination in the verbal probe while maintaining structured output integrity across all three families we tested.

C Prompt Templates

Each of the four channels described in Section 3.1 is driven by a dedicated prompt template. The environment-director prompt requests weather, event, and faction-tension updates as a JSON object validated against a pydantic schema. The npc-policy prompt requests per-group intent weights and target-priority lists on the clamped $[0, 3]$ scale. The strategic-plan prompt requests a long-form primary goal, a list of constraints, and a free-form summary intended as the verbal channel of the agent. The colony-agent prompt requests a per-step build plan following the Perceive–Plan–Ground–Execute–Evaluate–Reflect pattern, with each phase emitted as a separately validated JSON sub-object.

The four prompts share a common observation envelope that summarises recent world state, scenario blueprints, and the persistent world-summary memory into which anchors have been injected. Outputs are validated and clamped through a shared guardrail layer before they act on the world. Verbatim prompt templates and the pydantic response-schema definitions are released with the code package; we omit them from the paper body to preserve length, and refer the reader to the released repository for the exact strings used in the pilot.